Salt, Silver, and the Shape of Exchange: Overland Trade Networks of the Southern Baltic, 1350–1450

by

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Abstract

This dissertation reconstructs the overland trade networks that linked the salt works of Lüneburg, the silver markets of the Saxon uplands, and the Hanseatic ports of the southern Baltic between 1350 and 1450. Historians have long treated the Hanse as a maritime league whose inland connections were secondary. Drawing on 4,118 entries from eleven toll registers, together with merchant correspondence and municipal account books, I argue instead that the land routes formed a dense, resilient network whose structure shaped the fortunes of the coastal cities.

The study proceeds in three steps. Chapter One sets out the historiographical problem and the questions the dissertation answers. Chapter Two describes the sources, the procedure used to reconcile divergent units of account, and a gravity model of trade that estimates the friction imposed by distance and tolls. Chapter Three applies these methods to the salt roads, showing that a handful of inland toll towns acted as brokers between otherwise separate regional systems, and that the plague years of 1349–1351 and 1374 reorganised rather than destroyed this network.

The findings suggest that the prosperity of Lübeck rested as much on its control of overland brokerage as on its fleet, and that the much-discussed fifteenth-century decline of the Hanse began with a slow erosion of these inland positions.

For my parents,
who first walked the old salt road with me.

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CHAPTER ONE

Introduction

In the spring of 1412 a Lüneburg carter named Hinrik Vos paid four pennies at the toll bridge of Mölln for a wagon of salt bound for Lübeck. The entry is unremarkable; the same register records more than three hundred such payments that year. Yet entries of this kind, read in aggregate, describe something that the familiar narrative of the Hanseatic League has largely ignored: a web of overland routes, carriers and toll stations without which the maritime trade of the Baltic could not have functioned.

This dissertation reconstructs that web. It asks how goods moved between the inland centres of production and the coastal ports of the southern Baltic between 1350 and 1450, who controlled the points at which routes converged, and how the network responded to the demographic shocks of the late fourteenth century.

1.1 The Maritime Bias of Hanseatic History

The Hanse has been written about chiefly as a league of seaports. Its great nineteenth-century historians took their cue from the surviving records of the *Hansetage*, the diets at which delegates of the member towns negotiated privileges abroad, and these records are overwhelmingly concerned with ships, cargoes and foreign counting houses (Dollinger 1964; Hammel 1988). The resulting picture is one of a fleet with towns attached.[1]

More recent work has complicated this picture without overturning it. Studies of individual merchant firms show that the same families who freighted ships at Lübeck also financed wagon trains to Nuremberg and Kraków (Selzer 2003). Studies of the salt trade show that Lüneburg salt reached Scandinavia only after a journey

[1]The phrase is Brandt (1997)'s, who uses it to describe, and to criticise, the older consensus.

of some eighty kilometres overland (Witthöft 1976). But these findings remain scattered, and no study has yet treated the land routes as a system in their own right.

1.2 Argument

The central claim of this dissertation is that the overland routes of the southern Baltic formed a coherent network whose structure, rather than the volume of goods it carried, determined which towns prospered. Three subsidiary claims support it.

First, the network was *sparse but well connected*: most towns traded with only a few partners, yet any two towns could be linked through a short chain of intermediaries. Second, a small number of inland toll towns occupied brokerage positions, sitting on the only paths between otherwise separate regional systems. Third, the plague years did not destroy the network but reorganised it, concentrating traffic on fewer, better-maintained roads and thereby strengthening the brokers.

1.3 Plan of the Dissertation

Chapter 2 introduces the sources, chiefly eleven surviving toll registers, and the procedures used to make their entries comparable. It also sets out the two quantitative tools used throughout: a gravity model of bilateral trade and a measure of brokerage drawn from social network analysis. Chapter 3 applies these tools to the salt roads that connected Lüneburg to the Baltic coast, tracing how traffic and brokerage shifted across the century. Appendix A lists the registers and summarises their coverage.

CHAPTER TWO

Sources and Method

2.1 The Toll Registers

The core evidence for this study consists of eleven toll registers kept between 1352 and 1447 at bridges, gates and river crossings along the roads between Lüneburg, Hamburg, Lübeck and the Mecklenburg ports. Each register records, at minimum, the date, the toll paid and a description of the goods; most also name the carrier and the destination. Table 2.1 summarises the surviving series.

The registers were not kept for the historian's convenience. They served to record revenue owed to a lord or a town, and their categories reflect fiscal rather than commercial interests.[1] Goods that paid no toll, such as the possessions of clergy or the wares of citizens of privileged towns, are largely invisible.

[1] At Lauenburg, for example, wagons of herring and wagons of salt paid the same toll and are therefore recorded together until 1392, when the duke raised the rate on salt.

Table 2.1. Surviving toll registers used in this study

Station	Years covered	Entries	Named destinations
Mölln, bridge	1352–1447	1,204	38
Lauenburg, ferry	1361–1430	812	27
Oldesloe, gate	1374–1441	655	22
Ratzeburg, causeway	1380–1425	493	19
Boizenburg, crossing	1388–1446	441	17
Six minor stations	various	513	31
Total		4,118	74

2.2 Merchant Correspondence and Civic Accounts

The registers show what moved and where; they rarely show why. For motive and organisation I rely on two further bodies of evidence. The first is the correspondence of the Veckinchusen and Castorp firms, some 600 letters in which partners discuss prices, routes and the reliability of carriers (Selzer 2003). The second is the account books of the Lübeck and Lüneburg treasuries, which record payments for road repairs, escorts and the purchase of toll rights.

These sources are uneven. The letters survive in clusters that reflect the accidents of family archives, and the treasury accounts are complete only for Lübeck after 1380. They are used here to interpret patterns found in the registers rather than to establish those patterns independently.[2]

2.3 Reconciling Units of Account

The registers measure goods in wagons, barrels, *Last* and occasionally in value. To compare them, every entry was converted into an estimated weight in hundredweights. For salt, a Lüneburg barrel held about 140 kilograms, and a wagon carried between eight and twelve barrels (Witthöft 1976). Where a register records only the toll paid, the weight was estimated from the station's published tariff. The resulting series is approximate, but the errors are unlikely to vary systematically between routes, which is what matters for the comparisons below.

2.4 A Gravity Model of Overland Trade

To estimate how strongly distance and tolls discouraged trade, I fit a gravity model of the kind long used in economic history (Tinbergen 1962; Chilosi and Volckart 2013). The expected flow of goods F_{ij} between towns i and j is

$$F_{ij} = G \, \frac{M_i^{\alpha} M_j^{\beta}}{d_{ij}^{\gamma}} \, e^{-\delta \tau_{ij}}, \tag{2.1}$$

where M_i and M_j are proxies for the economic size of the two towns, d_{ij} is the road distance between them, and τ_{ij} is the number of toll stations on the cheapest path.

[2] Where the letters and the registers disagree, as they occasionally do about the volume of a particular shipment, the register is preferred, since its keeper had no reason to exaggerate.

Taking logarithms gives a linear model,

$$\ln F_{ij} = \ln G + \alpha \ln M_i + \beta \ln M_j - \gamma \ln d_{ij} - \delta \tau_{ij} + \varepsilon_{ij}, \tag{2.2}$$

which can be estimated by ordinary least squares on the 412 town pairs for which at least one flow is recorded. Town size is proxied by the number of households in the nearest surviving tax list.

2.5 Measuring Brokerage

The gravity model describes flows between pairs of towns. To describe positions within the network as a whole, I use betweenness centrality (Freeman 1977). For a town v, it is

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}, \tag{2.3}$$

where σ_{st} is the number of shortest paths between towns s and t and $\sigma_{st}(v)$ is the number of those paths that pass through v. A town with high betweenness sits on many of the routes that connect others; it is, in the language of network analysis, a broker (Padgett and Ansell 1993).

CHAPTER THREE

The Salt Roads

3.1 The Shape of the Network

Figure 3.1 shows the overland network reconstructed from the registers for the decade 1400–1409. Line weight is proportional to the estimated weight of goods carried. Two features stand out. The first is the dominance of the Lüneburg–Lübeck axis, which carried nearly half of all recorded salt. The second is the position of Mölln, through which every route from the salt works to the eastern Baltic ports had to pass.

The network is sparse. Of the 55 possible connections between its eleven principal towns, only twelve carry recorded traffic. Yet the mean shortest path between any two towns is just 2.4 steps, the hallmark of a network that is cheap to maintain and still well integrated.[1]

3.2 Distance, Tolls and Trade

Estimating Equation (2.2) on the full set of town pairs yields a distance elasticity of $\hat{\gamma} = 1.31$ and a toll coefficient of $\hat{\delta} = 0.22$. Each additional toll station thus reduced expected trade by roughly a fifth, holding distance constant. Table 3.1 reports the estimates for the two halves of the period.

The contrast between the two halves is the most important result of this chapter. After 1400, distance mattered less and tolls mattered more. Roads had improved, carts had grown larger, and the cost of moving a barrel one more mile had fallen. But the toll stations had multiplied and their rates had risen, so that the decisive cost of trade was increasingly a political one (Brandt 1997).

[1] By comparison, a random network with the same number of towns and connections would have a mean path length of about 2.9 steps.

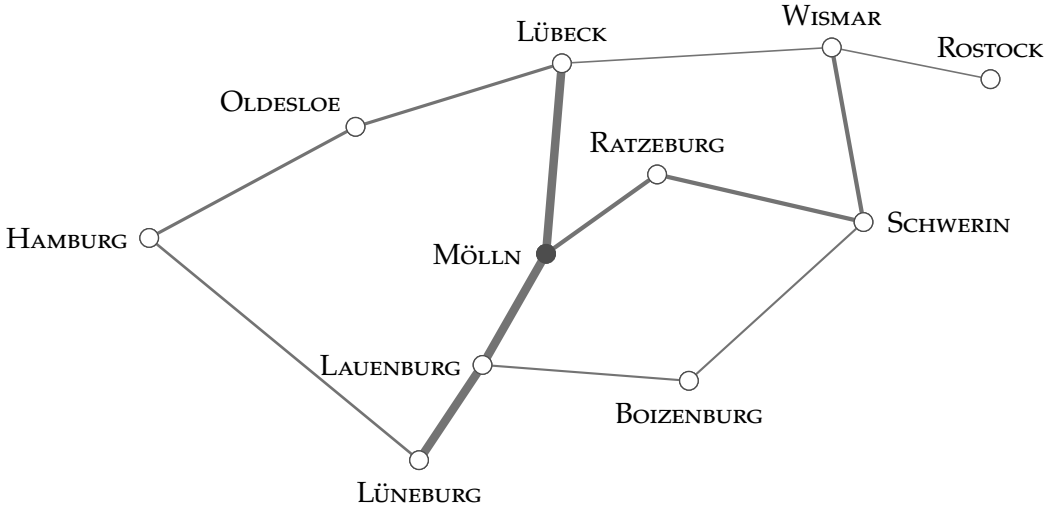


Figure 3.1. Overland routes of the southern Baltic, 1400–1409. Line weight is proportional to recorded tonnage; the filled node marks the principal broker.

Table 3.1. Gravity model estimates before and after 1400

	1350–1399	1400–1450	All years
Origin size, α	0.84	0.91	0.87
Destination size, β	0.72	0.79	0.76
Distance, γ	1.44	1.18	1.31
Tolls, δ	0.17	0.27	0.22
Town pairs	231	274	412
R^2	0.58	0.63	0.61

3.3 The Brokers

Betweenness centrality, computed with Equation (2.3), confirms what the map suggests. Mölln's score is more than twice that of any other town, and Lauenburg and Ratzeburg follow at a distance. None of these was a large place; Mölln had perhaps 1,500 inhabitants. Their importance lay entirely in their position.

Lübeck understood this. In 1359 the city acquired Mölln and its toll from the dukes of Saxe-Lauenburg by way of a mortgage that it never allowed to be redeemed (Hammel 1988). The purchase has usually been explained as a matter of security. The network evidence suggests a simpler motive: by owning the broker, Lübeck owned the route.

3.3.1 Reorganisation after the Plague

The plague of 1349–1351 and its recurrence in 1374 reduced the population of the region by perhaps a third. One might expect the network to have contracted in proportion. It did not. The number of active routes fell by only two, and the share of traffic passing through the three principal brokers rose from 54 to 68 per cent. Fewer carriers travelled on fewer roads, and those roads were the ones that the brokers controlled.

3.4 Silver and the Return Trade

Salt travelled north; silver travelled south. The registers record the return cargoes far less carefully, because coin and bullion in the hands of merchants were often exempt from toll, but the merchant letters fill much of the gap. Partners of the Castorp firm wrote of settling salt purchases at Lüneburg with Saxon silver brought up from the Erzgebirge by way of Magdeburg, and of carrying the surplus to Lübeck for shipment to Bergen and Novgorod (Selzer 2003).

The return trade matters for the argument because it explains why the brokers could raise their tolls without driving traffic away. A carter who brought salt to Lübeck could expect a paying cargo for the journey home, so that the cost of a toll was spread across two loads rather than one. The gravity estimates in Table 3.1, which treat each direction separately, therefore overstate the burden that tolls placed on any single round trip, though not the political leverage that they gave to those who collected them.[2]

[2] A model that pairs outbound and return flows would require matching individual carters across registers, which is possible for only 212 of the 4,118 entries.

3.5 Conclusion

The overland network of the southern Baltic was neither a minor appendage to the maritime trade nor a simple reflection of it. Its structure created positions of power that had little to do with size or wealth, and the cities that grasped this, above all Lübeck, turned that structure to their advantage. When, in the fifteenth century, territorial princes began to reclaim the inland toll stations, they struck at the foundation on which Hanseatic commerce had quietly rested.

APPENDIX A

The Toll Registers

This appendix describes the conventions used when transcribing the registers listed in Table 2.1.

A.1 Transcription

Entries were transcribed in full, preserving the original spelling of names and goods. Dates given by saints' days were converted to the Julian calendar using the standard tables. Where an entry is damaged, the legible portion is recorded and the remainder marked as missing; no entries were reconstructed by conjecture.

A.2 Classification of Goods

Goods were assigned to eleven categories: salt, herring, grain, beer, cloth, wax, furs, metals, timber, wine and miscellaneous. Mixed cargoes were assigned to the category named first in the entry, on the assumption, supported by the tariffs, that toll keepers listed the most valuable item first.

A.3 Estimating Weights

For salt, one barrel is taken as 140 kilograms and one wagon as ten barrels. For other goods, the conversion factors follow the tariffs of the respective stations. Entries recording only a payment were converted using

$$w = \frac{p}{r_{g}}, \tag{A.1}$$

where p is the toll paid in pennies and r_{g} is the station's rate per hundredweight for goods of category g.

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